ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 16	Sum of electrons in a neutral atom.
2	**(a)** 12	2, 8, 2.
3	**(d)** 35.5 u	26.25 + 9.25.
4	**(d)** 1	2, 8, 8, 1.
5	**(d)** 4	2, 8, 4.
6	**(d)** 10.8 u	2 + 8.8.
7	**(c)** calcium	Z = 20.
8	**(b)** aluminium	2, 8, 3.
9	**(d)** neutrons	Mass number differs.
10	**(b)** 2, 8	Loses 2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A = 24 (1); 2, 8, 2 (1).	1+1
12	(63 x 70 + 65 x 30)/100 (1); = 63.6 u (1).	1+1
13	Z = 24 - 12 = 12 (1); 2, 8, 2 (1).	1+1
14	Same electrons so same chemical behaviour (1); Different masses so different physical properties (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A = 40 (1); 2, 8, 8, 2; valency 2 (1); Calcium (1).	1+1+1
16	2, 8, 3 (1); Z = 13 (1); Valency 3 (1).	1+1+1
17	Electrons jump between shells (1); Photons of fixed energy are emitted or absorbed (1); Gives lines in the spectra (1).	1+1+1
	OR: Both: p = e = 17 (1); n = 18 and 20 (1); Isotopes (1).	1+1+1
18	Iodine-131 is absorbed by the thyroid, treats goitre (1); Cobalt-60 gives gamma rays (1); Used in cancer therapy (1).	1+1+1
	OR: Its only shell (K) is complete with 2 electrons (1); Stable, unreactive (1); Like other noble gases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	p = e = 13, n = 14 (1); 2, 8, 3 (1); Valency 3 (1); Al^{3+} (1); Al_2O_3 (1).	1+1+1+1+1
	OR: C-12: 6 p, 6 n (1); C-13: 7 n (1); C-14: 8 n, radioactive (1); Dating of fossils by decay of C-14 (1); Half life about 5730 years (1).	1+1+1+1+1
20	Let x per cent be Cl-35 (1); 35x + 37(100 - x) = 3550 (1); x = 75 (1); Cl-35 75 per cent, Cl-37 25 per cent (1); Mixture of isotopes gives non-integer mass (1).	1+1+1+1+1
	OR: Same electron arrangement (1); Chemical behaviour identical (1); Mass differs (1); Diffusion or centrifuge or mass spectrometer (1); Used to enrich uranium (1).	1+1+1+1+1

Mistakes to avoid:

- Confusing atomic number with mass number, or finding neutrons as $A + Z$ instead of $A - Z$.
- Filling the M shell with 18 electrons for elements up to calcium; the outer shell cannot hold more than 8, so Ca is 2, 8, 8, 2 and not 2, 8, 10.
- Taking the valency of an element like nitrogen (2, 5) as 5 instead of $8 - 5 = 3$.
- Saying that isotopes differ in the number of protons or electrons; they differ only in the number of neutrons.
- Assuming the atomic mass of chlorine is a whole number or that every chlorine atom weighs 35.5 u; it is a weighted average of isotopes.
- Attributing the discovery of the nucleus to Thomson, or the neutron to Rutherford (nucleus: Rutherford; neutron: Chadwick; electron: Thomson).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** carbon (2, 4)	4 for C; others 2, 1, 1.
2	**(a)** carbon dating	Radioactive dating.
3	**(a)** 2, 8, 8	Gains 2.
4	**(b)** 3	8 - 5.
5	**(a)** 18	Sum.
6	**(d)** 20	39 - 19.
7	**(a)** 6	Outer shell.
8	**(c)** 20	Calcium.
9	**(a)** 10	13 - 3.
10	**(a)** argon	2, 8, 8.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Neutral atom has 17 electrons (1); Atomic number = 17 (1).	1+1
12	Octet is stable (1); No need to gain or lose electrons (1).	1+1
13	p = e = 13 (1); n = 14 (1).	1+1
14	Mass number is a whole number count of nucleons (1); Atomic mass is the average relative mass in u (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Atom is neutral (1); Nucleus holds the protons (+) (1); Equal negative electrons balance (1).	1+1+1
16	Na loses 1 electron: 2, 8 (1); Cl gains 1: 2, 8, 8 (1); Both attain octets (1).	1+1+1
17	Z = protons (1); A = nucleons (1); Atomic mass is average in u (1).	1+1+1
	OR: Two isotopes 35 and 37 (1); Weighted average 3 : 1 (1); 35.5 u (1).	1+1+1
18	Most alpha particles pass through the foil (1); Nucleus is a tiny part of the atom (1); Atom is about 100000 times larger in size (1).	1+1+1
	OR: Protons = 20 (1); Calcium (1); Ca^{2+} (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Thomson (1); Rutherford (1); Bohr (1); Stability discussion (1); Drawbacks (1).	1+1+1+1+1
	OR: 2, 8, 8, 1 (1); Z = 19 (1); Valency 1 (1); Potassium (1); Soft, reactive metal (1).	1+1+1+1+1
20	Observations of alpha scattering (1); Conclusions (1); Ratio = 10^5 (1); Atom 100000 times the nucleus in diameter (1); Most space is empty (1).	1+1+1+1+1
	OR: Nucleons: protons and neutrons (1); Hydrogen nucleus 1 u (1); Electron 1/1836 u (1); Fraction = 1/1837, about 0.05 per cent (1); Hence mass is mostly in the nucleus (1).	1+1+1+1+1

Mistakes to avoid:

- Confusing atomic number with mass number, or finding neutrons as $A + Z$ instead of $A - Z$.
- Filling the M shell with 18 electrons for elements up to calcium; the outer shell cannot hold more than 8, so Ca is 2, 8, 8, 2 and not 2, 8, 10.
- Taking the valency of an element like nitrogen (2, 5) as 5 instead of $8 - 5 = 3$.
- Saying that isotopes differ in the number of protons or electrons; they differ only in the number of neutrons.
- Assuming the atomic mass of chlorine is a whole number or that every chlorine atom weighs 35.5 u; it is a weighted average of isotopes.
- Attributing the discovery of the nucleus to Thomson, or the neutron to Rutherford (nucleus: Rutherford; neutron: Chadwick; electron: Thomson).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $^{32}_{16}S$	S-32 has 16 neutrons.
2	**(b)** 18	17 + 1.
3	**(d)** 3	2, 8, 5.
4	**(c)** 2 neutrons	Same Z.
5	**(d)** Na^+	11 - 1.
6	**(a)** 1 to 3	Lose electrons.
7	**(c)** 25 u	Mean.
8	**(c)** F^- and Na^+	Both 10.
9	**(d)** 4 to 7	Gain electrons.
10	**(d)** deuterium and tritium	n = 1, 2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Particle in the nucleus: proton or neutron (1); 23 (1).	1+1
12	Na^+: 2, 8; Cl^-: 2, 8, 8 (1); Noble gas arrangements (octet) (1).	1+1
13	Electron mass is about 1/1836 u (1); Nucleons make nearly all the mass (1).	1+1
14	Nucleus about 10^-15 m (1); Atom about 10^-10 m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Protons = 20 (1); Neutrons = 20 (1); Electrons = 18 (1).	1+1+1
16	Y loses 2, X gains 1 (1); YX_2 (1); Ionic bond (1).	1+1+1
17	18, 19, 20 protons (1); Neutrons 22, 21, 20 (1); Same mass number 40 (1).	1+1+1
	OR: p = 17, n = 18 (1); Z = 17 (1); 2, 8, 7 (1).	1+1+1
18	N 5, valency 3 (1); F 7, valency 1 (1); Al 3, 3; Si 4, 4 (1).	1+1+1
	OR: Accelerating electron should radiate energy and spiral in (1); Atoms would not be stable (1); Bohr's shells solved it (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	1, 2, 3 valence electrons: metals (1); 5, 6, 7: non-metals (1); Valency = valence electrons or 8 - valence electrons (1); Na: Na^+; Cl: Cl^- (1); Mg: Mg^{2+}; O: O^{2-} (1).	1+1+1+1+1
	OR: Mg: 2, 8, 2 loses 2; O gains 2 (1); Na loses 1; Cl gains 1 (1); Higher charges in MgO (1); Stronger attraction (1); Higher melting point (1).	1+1+1+1+1
20	O: 8, 8, 8; 2, 6 (1); O2-: 8, 8, 10; 2, 8 (1); Al: 13, 14, 13; 2, 8, 3 (1); Al3+: 13, 14, 10; 2, 8 (1); Compare neon-like ions (1).	1+1+1+1+1
	OR: (20 x 90 + 21 x 1 + 22 x 9)/100 (1); = (1800 + 21 + 198)/100 (1); = 20.19 u (1); Neon (1); Isotopes Ne-20, Ne-21, Ne-22 (1).	1+1+1+1+1

Mistakes to avoid:

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- Taking the valency of an element like nitrogen (2, 5) as 5 instead of $8 - 5 = 3$.
- Saying that isotopes differ in the number of protons or electrons; they differ only in the number of neutrons.
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- Attributing the discovery of the nucleus to Thomson, or the neutron to Rutherford (nucleus: Rutherford; neutron: Chadwick; electron: Thomson).

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